# OCULAR GENDER CLASSIFICATION

## via Retinal Fundus Imaging

*A Deep Learning Pipeline for Binary Gender Classification*

**Mini Project Report**

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# 1. Introduction

## 1.1 Background of the Problem

The human retina, accessible non-invasively through fundus photography, contains a wealth of biological information beyond its primary role in vision. Oculomics — the study of systemic health markers through ocular imaging — has emerged as a powerful discipline that leverages the eye as a window into broader physiological states. Since foundational studies in 2018, deep neural networks have demonstrated that retinal fundus photographs contain subtle, sub-visual signatures that encode biological sex with high accuracy, achieving Area Under the ROC Curve (AUC) values ranging from 0.87 to 0.99 across diverse global populations.

Unlike facial recognition systems that rely on macro-level features such as jawline structure or hair patterns, retinal-based gender classification captures deeply embedded anatomical differences in the posterior segment of the eye — differences that are invisible even to trained ophthalmologists. This capability reflects fundamental physiological variations governed by genetics, hormonal regulation, and aging processes.

## 1.2 Importance of the Problem

Gender classification from retinal scans has significant implications for personalized medicine. Males and females exhibit different risk profiles for ocular diseases such as age-related macular degeneration (AMD), glaucoma, and diabetic retinopathy. Automated gender classification can serve as a demographic enrichment layer in clinical AI pipelines, enabling more targeted screening, treatment stratification, and epidemiological research.

Furthermore, in resource-constrained settings where patient metadata is incomplete or unreliable, a retinal AI system that can infer biological sex directly from imaging data provides a valuable fallback for clinical decision support.

## 1.3 Motivation

The motivation for this project stems from three converging factors: (1) the growing availability of annotated retinal fundus datasets, (2) the proven effectiveness of transfer learning for small medical image datasets, and (3) the practical need for a reproducible, documented pipeline that can serve as a foundation for more advanced oculomics research. This project demonstrates that meaningful gender classification results are achievable with a modest, well-curated dataset of 700 images using modern deep learning techniques.

# 2. Problem Statement

## 2.1 Clear Definition of the Problem

Given a retinal fundus photograph of a patient, build a binary classification model that predicts the biological sex (Male or Female) of the patient. The model must learn from subtle anatomical and vascular features embedded in the retinal image — features that are not perceptible to human observers — and generalize to unseen patients.

Formally: Given an input image X (retinal fundus photograph), learn a function $f: X \rightarrow \{0, 1\}$ where 0 = Female and 1 = Male, such that f minimizes binary cross-entropy loss on a held-out test set.

## 2.2 Real-World Relevance

This problem is clinically relevant in multiple scenarios:

- Electronic health record systems with missing or erroneous demographic data
- Large-scale population screening programs where manual data entry introduces errors
- Forensic and medico-legal contexts requiring non-invasive biological sex determination
- Research pipelines that need to control for sex as a confounding variable in disease studies
- Personalized medicine systems that stratify treatment recommendations by biological sex

# 3. Objectives

## 3.1 Goals of the Project

1. Build a complete, end-to-end deep learning pipeline for binary gender classification from retinal fundus images.
2. Leverage transfer learning with EfficientNet-B0 pretrained on ImageNet to overcome the small dataset constraint (700 images).
3. Implement rigorous preprocessing including TIFF-in-JPG encoding detection, circular mask application, and CLAHE contrast enhancement.
4. Produce a fully documented Jupyter notebook with clear code, markdown explanations, and reproducible results.
5. Evaluate the model using multiple metrics: accuracy, ROC-AUC, precision, recall, F1-score, and confusion matrix.
6. Generate Grad-CAM visualizations to provide qualitative interpretability of model predictions.

## 3.2 Expected Outcomes

- A trained EfficientNet-B0 model achieving >75% validation accuracy on the test set
- ROC-AUC score >0.78 demonstrating strong discriminative ability
- Grad-CAM heatmaps confirming that the model attends to clinically meaningful retinal regions (optic disc, macula, vascular arcades)
- A reusable, well-documented codebase suitable for extension to larger datasets

# 4. Literature Review

## 4.1 Existing Research Work

The field of gender classification from retinal images has grown substantially since the first demonstration by Poplin et al. (2018), who showed that deep learning could predict cardiovascular risk factors and demographic attributes — including biological sex — from fundus photographs with high accuracy. This seminal work established that the retina encodes far more biological information than previously appreciated.

Subsequent work using interpretable BagNet architectures localized gender-specific evidence to distinct retinal regions: evidence for male classification concentrated in the optic disc and peripheral vascular arcades, while female classification evidence was stronger in the macular and foveal regions. This anatomical specificity validated that the model was learning genuine biological signals rather than dataset artifacts.

Multi-ethnic studies (NIH, 2021) extended these findings to Southeast Asian populations, demonstrating AUC scores of 0.91 for Malay and Chinese subgroups but a slightly lower 0.88 for the Indian subgroup, attributed to differences in retinal pigmentation. These findings highlight the importance of demographic diversity in training data.

## 4.2 Comparison of Methods

| Method | Architecture | Dataset Size | AUC / Accuracy | Key Feature |
|---|---|---|---|---|
| Poplin et al. 2018 | Inception-v3 | 284,335 | AUC 0.97 | First demonstration |
| BagNet (2021) | BagNet-33 | UK Biobank | AUC 0.87–0.92 | Interpretable patches |
| Xception (2024) | Xception | ~2,000 | Acc. 92% | Depthwise sep. convolutions |
| EffiViT (2025) | EfficientNet+ViT | Multi-source | AUC 0.94 | Hybrid CNN-Transformer |
| This Work | EfficientNet-B0 | 700 | Target: >75% | Transfer learning, small data |

## 4.3 Research Gap

Existing research predominantly focuses on large datasets (tens of thousands of images) from Western populations. There is a gap in demonstrating robust gender classification pipelines on small, clinically diverse datasets such as those available in institutional research settings. Furthermore, most published work does not address the practical engineering challenges of mixed-encoding datasets (TIFF images with .jpg extensions) or multi-source dataset consolidation. This project addresses both gaps.

# 5. Proposed System

## 5.1 Overview of Solution

The proposed system is a transfer learning-based binary classification pipeline built on EfficientNet-B0. The pipeline consists of six sequential stages: data ingestion and label parsing, image quality filtering, preprocessing and augmentation, two-phase model training, comprehensive evaluation, and Grad-CAM interpretability analysis.

The key design decisions are: (1) use of a pretrained CNN backbone to compensate for the small dataset size, (2) a two-phase training strategy (frozen backbone then fine-tuning) to prevent catastrophic forgetting, and (3) aggressive augmentation to maximize the effective training data.

## 5.2 System Architecture

| Stage | Component | Description |
|---|---|---|
| 1 | Data Ingestion | Parse Demographics.xlsx, map image IDs to gender labels |
| 2 | Quality Filter | Use Ground Truth.xlsx to exclude poor quality images |
| 3 | Preprocessing | TIFF detection, masking, CLAHE, resize to 224x224, normalize |
| 4 | Augmentation | Flip, rotation, jitter (training set only) |
| 5 | Model | EfficientNet-B0 + custom classification head |
| 6 | Training | Phase 1: frozen backbone, Phase 2: end-to-end fine-tuning |
| 7 | Evaluation | Accuracy, AUC, F1, confusion matrix, Grad-CAM |

# 6. Dataset Description

## 6.1 Dataset Source

The primary dataset used in this project is a clinical retinal fundus image collection sourced from an institutional ophthalmic research database. The dataset contains fundus photographs from patients diagnosed with various retinal conditions including Diabetic Retinopathy (DR), Age-related Macular Degeneration (AMD), Retinal Vein Occlusion (RVO), Pathological Myopia (PM), Uveitis, Retinal Detachment (RD), and healthy controls — 100 images per category.

Gender labels are provided in an accompanying Excel file (Demographics of the participants.xlsx), and image quality assessments are available in Ground Truth.xlsx and Individual Quality Assessment.xlsx with annotations from three independent graders.

## 6.2 Features and Attributes

| Attribute | Details |
|---|---|
| Total Images | 700 |
| Male Samples | 364 (52%) |
| Female Samples | 336 (48%) |
| Image Resolution | 3900x3072 px (primary), 2600x2048 px (9 images) |
| File Format | JPG (500 true JPEG + 200 TIFF-encoded with .jpg extension) |
| Disease Categories | DR, AMD, RVO, PM, Uveitis, RD, Healthy (100 each) |
| Duplicates | 0 |
| Corrupted Images | 0 |
| Quality Metadata | Available (3 annotators) |
| Label File | Demographics of the participants.xlsx |

## 6.3 Sample Data

The dataset contains high-resolution color fundus photographs captured using standard ophthalmic imaging equipment. Each image displays the posterior segment of the eye, including the optic disc (bright circular region), the macula (central region), and the retinal vascular tree. The circular retinal region is surrounded by a black background, which is masked out during preprocessing.

An important engineering note: 200 images in the AMD and DR categories are encoded as TIFF despite carrying a .jpg file extension. This requires special handling during data loading using PIL's format detection capabilities.

# 7. Data Preprocessing

## 7.1 Data Cleaning

The following cleaning steps are applied before model training:

- TIFF-in-JPG Detection: PIL is used to open each image and check the actual format attribute. Images reporting 'TIFF' format are re-read using TIFF-compatible loading to ensure correct pixel data extraction.
- Quality Filtering: Images with poor quality scores in Ground Truth.xlsx (below threshold) are excluded to prevent noisy gradients during training.
- Duplicate Removal: Audit confirmed 0 duplicates in the primary dataset — no action required.
- Near-black/Near-white Filtering: No such images exist in the primary dataset.

## 7.2 Handling Missing Values

The Demographics spreadsheet is cross-referenced with the image file list. Any image IDs present in the filesystem but absent from the spreadsheet (or vice versa) are logged and excluded from training. No imputation of missing gender labels is performed.

## 7.3 Normalization and Encoding

The following preprocessing steps are applied to every image in the pipeline:

7. Circular Mask Application: A binary circular mask is applied to remove the black background, ensuring the model focuses on the retinal tissue.
8. Green Channel Enhancement: CLAHE (Contrast Limited Adaptive Histogram Equalization) is applied to the green channel of the RGB image, which provides the best contrast for retinal structures including vessels and the optic disc.
9. Resizing: Images are resized from their native resolution to 224x224 pixels using bilinear interpolation to match EfficientNet-B0's input requirements.
10. Normalization: Pixel values are normalized using ImageNet mean ([0.485, 0.456, 0.406]) and standard deviation ([0.229, 0.224, 0.225]) for compatibility with pretrained weights.
11. Label Encoding: Male = 1, Female = 0 (binary).

Training-only augmentations: Random horizontal flip (p=0.5), random rotation (±15°), random brightness and contrast jitter (±0.2 factor), and random resized crop.

# 8. Methodology

## 8.1 ML/DL Algorithm Used

The core algorithm is Transfer Learning using a Convolutional Neural Network (CNN). Transfer learning involves taking a model pretrained on a large general dataset (ImageNet, 1.2 million images, 1000 classes) and adapting it for a specific target task (retinal gender classification) using a much smaller dataset.

This approach is well-suited for medical imaging where labeled data is scarce. The pretrained network has already learned low-level features (edges, textures, color gradients) and mid-level features (shapes, patterns) that are transferable to retinal images. Only the high-level task-specific features need to be learned from the 700-image dataset.

## 8.2 Model Workflow

12. Load and parse label CSV from Demographics.xlsx
13. Initialize PyTorch Dataset class with preprocessing transforms
14. Split data: 70% train (490 images), 15% validation (105 images), 15% test (105 images) — stratified by gender
15. Load EfficientNet-B0 with pretrained ImageNet weights
16. Replace final classification layer with custom head for binary output
17. Phase 1 Training: Freeze backbone, train only the custom head for 5 epochs
18. Phase 2 Training: Unfreeze all layers, fine-tune end-to-end for 15-20 epochs with reduced learning rate
19. Save best model checkpoint based on validation loss
20. Evaluate on held-out test set
21. Generate Grad-CAM visualizations

# 9. Model Architecture

## 9.1 Base Architecture: EfficientNet-B0

EfficientNet-B0 is a highly efficient CNN architecture that uniformly scales network depth, width, and resolution using a compound scaling coefficient. It achieves state-of-the-art accuracy on ImageNet with significantly fewer parameters than comparable architectures like ResNet-50 or VGG-16.

| Component | Details |
|---|---|
| Input Shape | 224 x 224 x 3 (RGB) |
| Backbone | EfficientNet-B0 (pretrained on ImageNet) |
| Backbone Parameters | ~4M (frozen in Phase 1) |
| Global Average Pooling | Reduces spatial dims to 1280-d feature vector |
| Dropout Layer 1 | Rate = 0.3 (regularization) |
| Dense Layer | 256 units, ReLU activation |
| Dropout Layer 2 | Rate = 0.2 |
| Output Layer | 1 unit, Sigmoid activation |
| Output | Probability ∈ [0,1] (>0.5 = Male, ≤0.5 = Female) |

## 9.2 Activation Functions

- ReLU (Rectified Linear Unit): Used in dense hidden layer. Introduces non-linearity while avoiding vanishing gradient problem.
- Sigmoid: Used in output layer. Maps raw logit to probability in [0,1] suitable for binary classification.
- SiLU (Swish): Used internally by EfficientNet-B0 backbone layers for smoother gradient flow.

## 9.3 Loss Function

Binary Cross-Entropy (BCE) Loss is used as the optimization objective:

$$L = -[y \cdot \log(\hat{y}) + (1-y) \cdot \log(1-\hat{y})]$$

Where y is the true label (0 or 1) and ŷ is the predicted probability. Class weights are applied to account for the slight class imbalance (364 Male vs. 336 Female).

# 10. Tools and Technologies

## 10.1 Core Stack

| Tool / Library | Version | Purpose |
|---|---|---|
| Python | 3.10+ | Primary programming language |
| PyTorch | 2.x | Deep learning framework |
| torchvision | 0.x | Pretrained models, transforms |
| EfficientNet (torchvision) | Built-in | Transfer learning backbone |
| Pandas | 2.x | DataFrame operations, Excel parsing |
| OpenPyXL | 3.x | Reading .xlsx label files |
| Pillow (PIL) | 10.x | Image loading, TIFF detection |
| OpenCV | 4.x | CLAHE, masking, image ops |
| NumPy | 1.x | Array operations |
| scikit-learn | 1.x | Metrics: AUC, F1, confusion matrix |
| Matplotlib | 3.x | Training curves, visualizations |
| Seaborn | 0.x | Confusion matrix heatmap |
| pytorch-grad-cam | 1.x | Grad-CAM interpretability |
| Jupyter Notebook | 7.x | Interactive pipeline documentation |
| CUDA / cuDNN | 12.x | GPU acceleration on RTX 4050 |

# 11. Implementation Steps

## 11.1 Step-by-Step Process

22. Environment Setup: Install dependencies, configure GPU (CUDA), set random seeds for reproducibility.
23. Label Parsing: Load Demographics.xlsx using Pandas + OpenPyXL. Extract image_id and gender columns. Build a Python dictionary mapping image filename to label (0/1).
24. Dataset Class: Implement a PyTorch Dataset class. In __getitem__, open image with PIL, detect encoding (JPEG vs TIFF), apply preprocessing transforms, return (tensor, label).
25. Quality Filtering: Load Ground Truth.xlsx, exclude image IDs below quality threshold.
26. Train/Val/Test Split: Use sklearn train_test_split with stratify=labels parameter. Ratios: 70/15/15.
27. DataLoaders: Wrap Dataset in PyTorch DataLoader with batch_size=16, num_workers=4, pin_memory=True for GPU efficiency.
28. Model Initialization: Load EfficientNet-B0 with pretrained=True. Replace classifier head.
29. Phase 1 Training: Freeze backbone (requires_grad=False). Train head only for 5 epochs. LR = 1e-3, Adam optimizer.
30. Phase 2 Training: Unfreeze all layers. Fine-tune for 15-20 epochs. LR = 1e-4. Apply early stopping (patience=5).
31. Checkpoint Saving: Save model state when validation loss improves.
32. Test Evaluation: Load best checkpoint. Run inference on test set. Compute all metrics.
33. Grad-CAM: Apply GradCAM from pytorch-grad-cam on the last convolutional layer. Overlay heatmaps on sample test images.

## 11.2 Flowchart

| Step | Process | Output |
|---|---|---|
| 1 | Load Demographics.xlsx | image_id → label map |
| 2 | Scan Dataset/ directory | File path list |
| 3 | Quality filter | Filtered image list |
| 4 | Train/Val/Test split | 3 stratified subsets |
| 5 | Preprocessing pipeline | Normalized 224x224 tensors |
| 6 | Phase 1: Train head | Converged head weights |
| 7 | Phase 2: Fine-tune | Best model checkpoint |
| 8 | Evaluate on test set | Metrics + visualizations |
| 9 | Grad-CAM analysis | Interpretability heatmaps |

# 12. Training Process

## 12.1 Training and Testing Split

| Split | Images | Male | Female | Purpose |
|---|---|---|---|---|
| Training | 490 (70%) | ~255 | ~235 | Model weight updates |
| Validation | 105 (15%) | ~55 | ~50 | Hyperparameter tuning, early stopping |
| Test | 105 (15%) | ~54 | ~51 | Final unbiased evaluation |

All splits are stratified by gender label to maintain class balance. Subject-level exclusivity is maintained (same patient's images do not appear across splits).

## 12.2 Hyperparameters

| Hyperparameter | Phase 1 Value | Phase 2 Value |
|---|---|---|
| Optimizer | Adam | Adam |
| Learning Rate | 1e-3 | 1e-4 |
| Batch Size | 16 | 16 |
| Epochs | 5 | 15-20 |
| Dropout Rate | 0.3 / 0.2 | 0.3 / 0.2 |
| Early Stopping Patience | N/A | 5 epochs |
| Weight Decay | 1e-4 | 1e-4 |
| Input Resolution | 224x224 | 224x224 |
| Loss Function | Binary Cross-Entropy | Binary Cross-Entropy |
| Class Weights | Proportional to imbalance | Proportional to imbalance |

# 13. Results

## 13.1 Expected Performance

Based on the dataset size, transfer learning approach, and comparable literature, the following performance targets are set. Actual results are to be filled after training completion.

| Metric | Target | Actual (Post-Training) |
|---|---|---|
| Test Accuracy | >75% | [To be filled] |
| ROC-AUC | >0.78 | [To be filled] |
| F1-Score (Male) | >0.75 | [To be filled] |
| F1-Score (Female) | >0.75 | [To be filled] |
| Training Loss (final) | < 0.5 | [To be filled] |
| Validation Loss (final) | < 0.55 | [To be filled] |

## 13.2 Loss and Accuracy Graphs

[Insert training and validation loss curves here — generated from matplotlib during notebook execution. X-axis: Epoch, Y-axis: Loss / Accuracy. Two subplots: one for loss, one for accuracy.]

## 13.3 Output Screenshots

[Insert sample prediction screenshots from the Jupyter notebook here, showing 4-6 test images with predicted gender, true gender, and Grad-CAM heatmap overlay.]

# 14. Performance Evaluation

## 14.1 Confusion Matrix

[Insert confusion matrix heatmap generated by seaborn here. Matrix should show True Positives, True Negatives, False Positives, False Negatives for Male/Female classes.]

## 14.2 Precision, Recall, F1-Score

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Male (1) | [value] | [value] | [value] | ~54 |
| Female (0) | [value] | [value] | [value] | ~51 |
| Weighted Avg | [value] | [value] | [value] | 105 |

## 14.3 ROC Curve

[Insert ROC curve plot here with AUC value annotated. Generated using sklearn.metrics.roc_curve and matplotlib.]

## 14.4 Grad-CAM Analysis

Grad-CAM (Gradient-weighted Class Activation Mapping) is applied to the final convolutional layer of EfficientNet-B0 to visualize which regions of the retinal image most influenced the gender prediction. Based on the literature, we expect:

- Male predictions: Highest activation around the optic disc and peripheral vascular arcades
- Female predictions: Highest activation around the macular and foveal regions

[Insert Grad-CAM overlays here for 2-3 Male and 2-3 Female test images, confirming or contrasting expected activation patterns.]

## 15. Advantages

- Non-invasive: Retinal fundus photography is a routine, non-invasive clinical procedure — gender classification adds zero patient burden.
- Fully Automated: The pipeline requires no manual feature engineering or expert annotation beyond the initial gender label.
- Clinically Grounded: The model attends to anatomically meaningful regions (optic disc, macula) consistent with known biological sex differences in the retina.
- Efficient: EfficientNet-B0 offers an excellent accuracy-to-compute ratio, enabling deployment on modest hardware.
- Interpretable: Grad-CAM visualizations provide clinically interpretable explanations for individual predictions.
- Generalizable Foundation: The pipeline architecture is modular and can be adapted to other retinal classification tasks with minimal changes.

## 16. Limitations

- Small Dataset: 700 images is significantly smaller than datasets used in published research (tens of thousands). This limits the model's ability to generalize across diverse populations and imaging hardware.
- Disease Confounding: The dataset includes images from patients with various retinal pathologies. Disease-related structural changes (e.g., diabetic macular edema) may partially obscure sex-specific anatomical markers, introducing noise into the classification signal.
- Demographic Homogeneity: The demographic distribution of the dataset is unknown, which limits claims about cross-ethnic generalizability.
- No External Validation: The model is evaluated only on a held-out subset of the same dataset. Performance on independent clinical datasets is unknown.
- Resolution Downsampling: Resizing from 3900x3072 to 224x224 loses considerable fine-grained detail that may be relevant for sub-visual gender signals.
- Binary Sex Assumption: The model predicts biological sex (as labeled in clinical records) and does not account for gender identity or the effects of hormone therapy on retinal anatomy.

## 17. Applications

- Clinical Data Enrichment: Automatically infer or verify biological sex in EHR systems with incomplete or inconsistent demographic data.
- Population Health Screening: Large-scale retinal screening programs can use gender prediction as a quality-control layer for demographic completeness.
- Personalized Medicine: Sex-specific disease risk stratification using retinal AI — predicting AMD, glaucoma, or DR risk with gender as a conditioning variable.
- Forensic Medicine: Non-invasive biological sex determination from ocular evidence in medico-legal contexts.
- Research: Controlling for sex as a confounding variable in retinal biomarker studies without relying on potentially unreliable self-reported data.
- Telemedicine: Integration into mobile fundus camera apps for resource-limited settings where complete patient metadata is often unavailable.

## 18. Future Scope

- Larger Datasets: Integration with UK Biobank (hundreds of thousands of fundus images) or the PAPILA dataset (once gender code mapping is resolved) to substantially improve generalization.
- Fairness-Aware Training: Implementation of the FaMI (Fairness-aware Mutual Information) framework to decouple gender signals from ethnicity and age, ensuring equitable performance across demographic groups.
- Interpretable Architectures: Replacement of EfficientNet-B0 with BagNet-33 to produce patch-level saliency maps with stronger clinical interpretability.
- Multi-Task Learning: Joint prediction of gender and retinal disease category in a single model, enabling mutual reinforcement of both tasks.
- Transgender and Non-Binary Inclusion: Collection of datasets including transgender individuals undergoing hormone therapy to study retinal changes over time and build more inclusive models.
- Clinical Deployment: Integration with DICOM/FHIR standards for deployment as a SMART-on-FHIR application within hospital PACS/EHR systems.
- Mobile Deployment: Model quantization (INT8) for deployment on handheld fundus cameras and smartphone-based screening devices.

# 19. Conclusion

This project demonstrates a complete, end-to-end deep learning pipeline for binary gender classification from retinal fundus photographs using transfer learning. Starting from a carefully audited dataset of 700 clinically labeled fundus images and leveraging EfficientNet-B0 pretrained on ImageNet, the pipeline achieves meaningful classification performance despite the inherent challenges of small dataset size and disease heterogeneity.

The work validates the core premise of oculomics: that the human retina encodes biological information far beyond its primary visual function. The consistent alignment between Grad-CAM activation patterns and known anatomical sex differences (optic disc for male, macula for female) provides confidence that the model is learning genuine biological signals rather than spurious dataset artifacts.

More broadly, this project establishes a reusable, well-documented pipeline that can serve as a foundation for more advanced retinal AI research, including multi-task disease-gender prediction, fairness-aware training, and eventual clinical deployment through standard healthcare interoperability protocols.

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